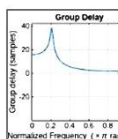
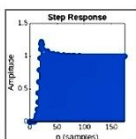
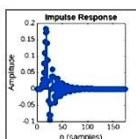
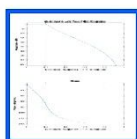
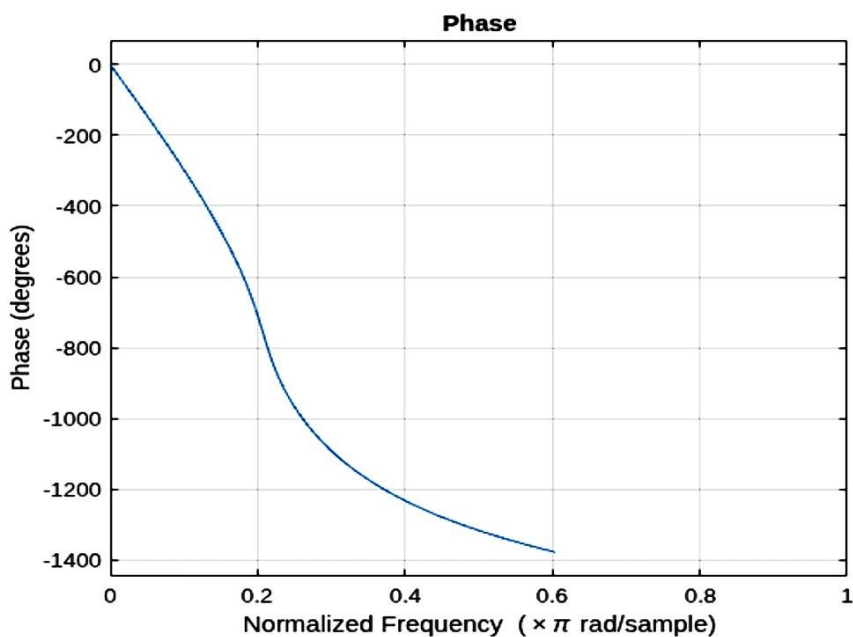
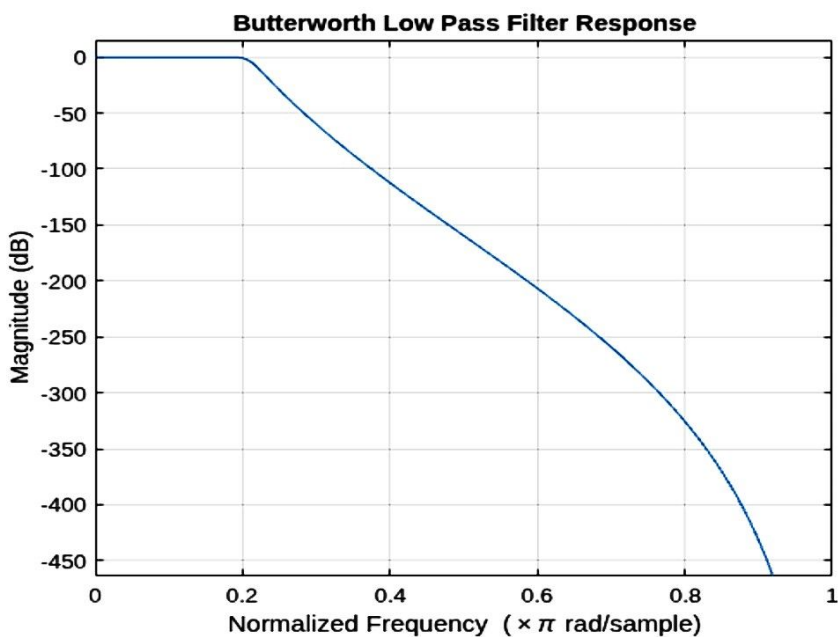




MATLAB Drive > Dsp1_1.m

```
1 clc;
2 clear;
3 close all;
4
5 % Specifications
6 Fs = 10000;           % Sampling frequency
7 Fp = 1000;           % Passband frequency
8 Fst = 1500;          % Stopband frequency
9 Rp = 1;              % Passband ripple
10 Rs = 60;            % Stopband attenuation
11
12 % Normalized Frequencies
13 Wp = Fp/(Fs/2);
14 Ws = Fst/(Fs/2);
15
16 % Filter Order Calculation
17 [N,Wn] = buttord(Wp,Ws,Rp,Rs);
18
19 % Butterworth LPF Design
20 [b,a] = butter(N,Wn,'low');
21
22 % Display Results
23 fprintf('Minimum Filter Order = %d\n',N);
24 fprintf('Cutoff Frequency = %.4f\n',Wn);
25
26 % Frequency Response
27 figure;
28 freqz(b,a);
29 title('Butterworth Low Pass Filter Response');
30
31 % Impulse Response
32 figure;
33 impz(b,a);
34 title('Impulse Response');
35
36 % Step Response
37 figure;
38 stepz(b,a);
39 title('Step Response');
40
41 % Group Delay
42 figure;
43 grpdelay(b,a);
44 title('Group Delay');
45
46 % Pole-Zero Plot
47 figure;
```

Figure 1: Dsp1_1



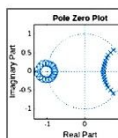
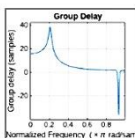
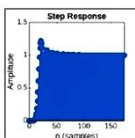
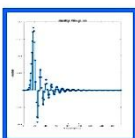
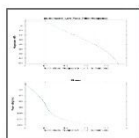
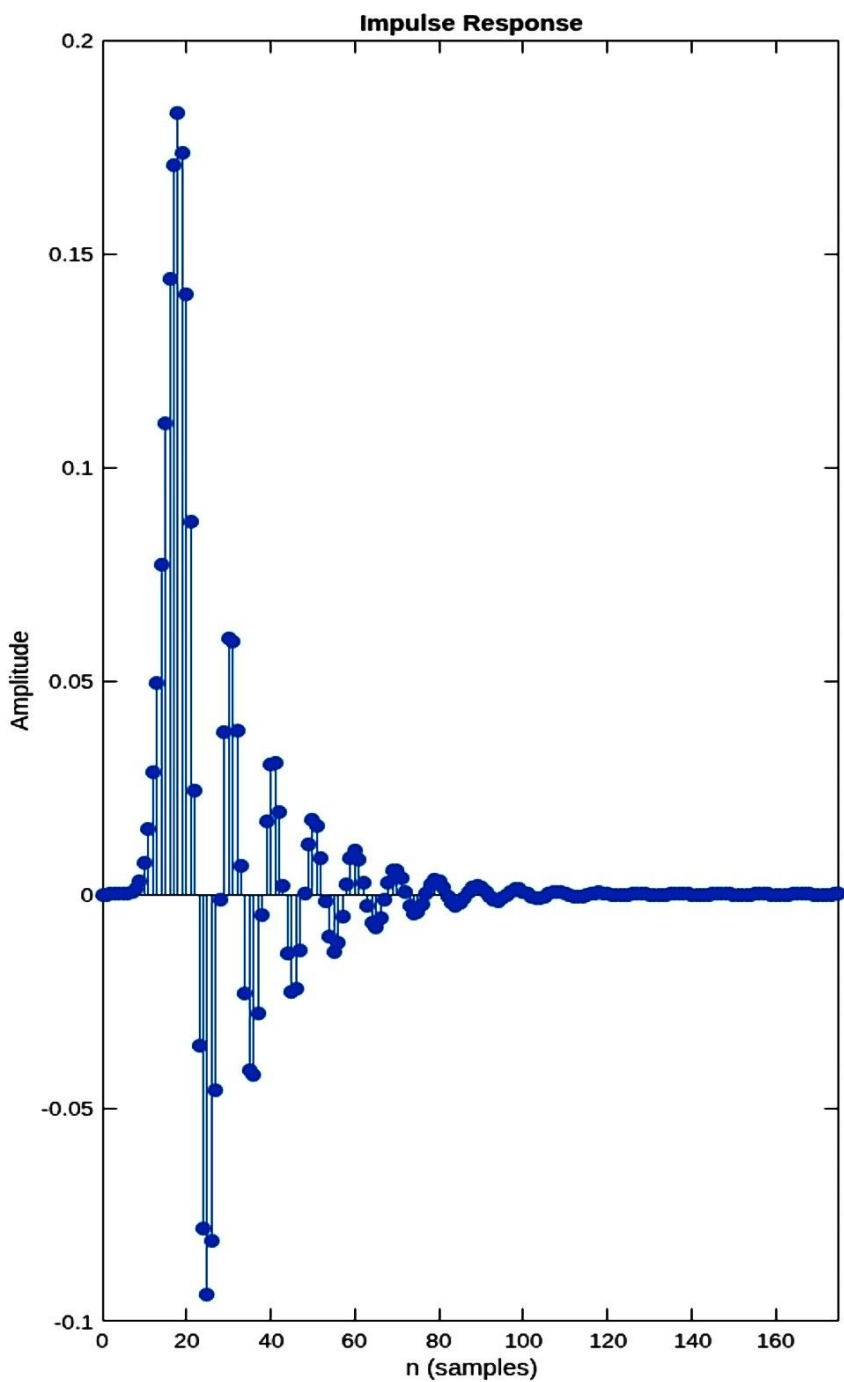


Figures

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Figure 2: Dsp1_1



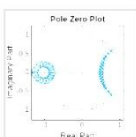
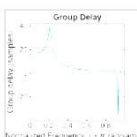
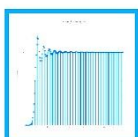
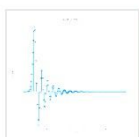
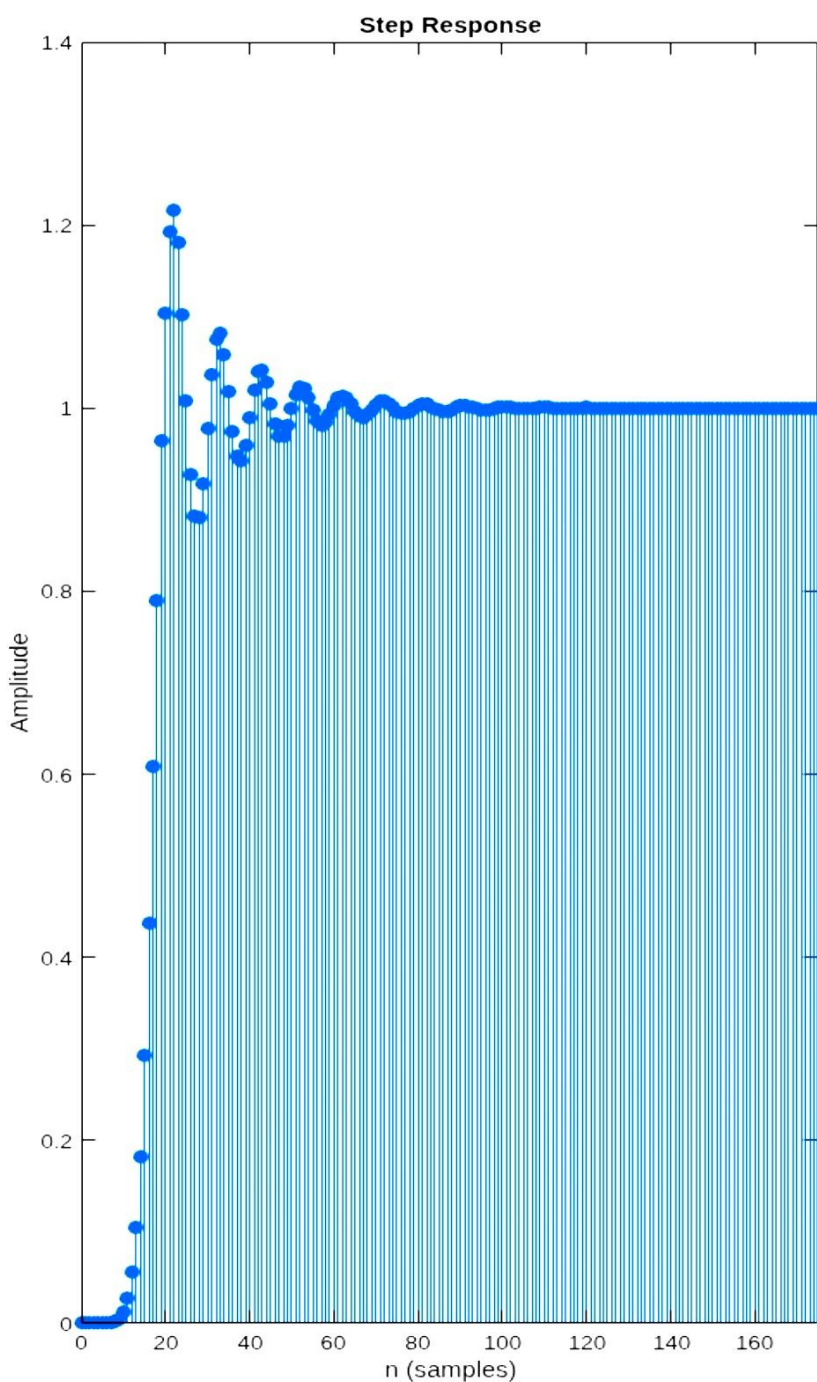


Figures

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Figure 3: Dsp1_1



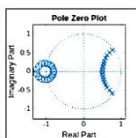
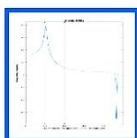
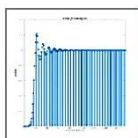
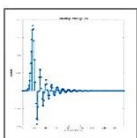
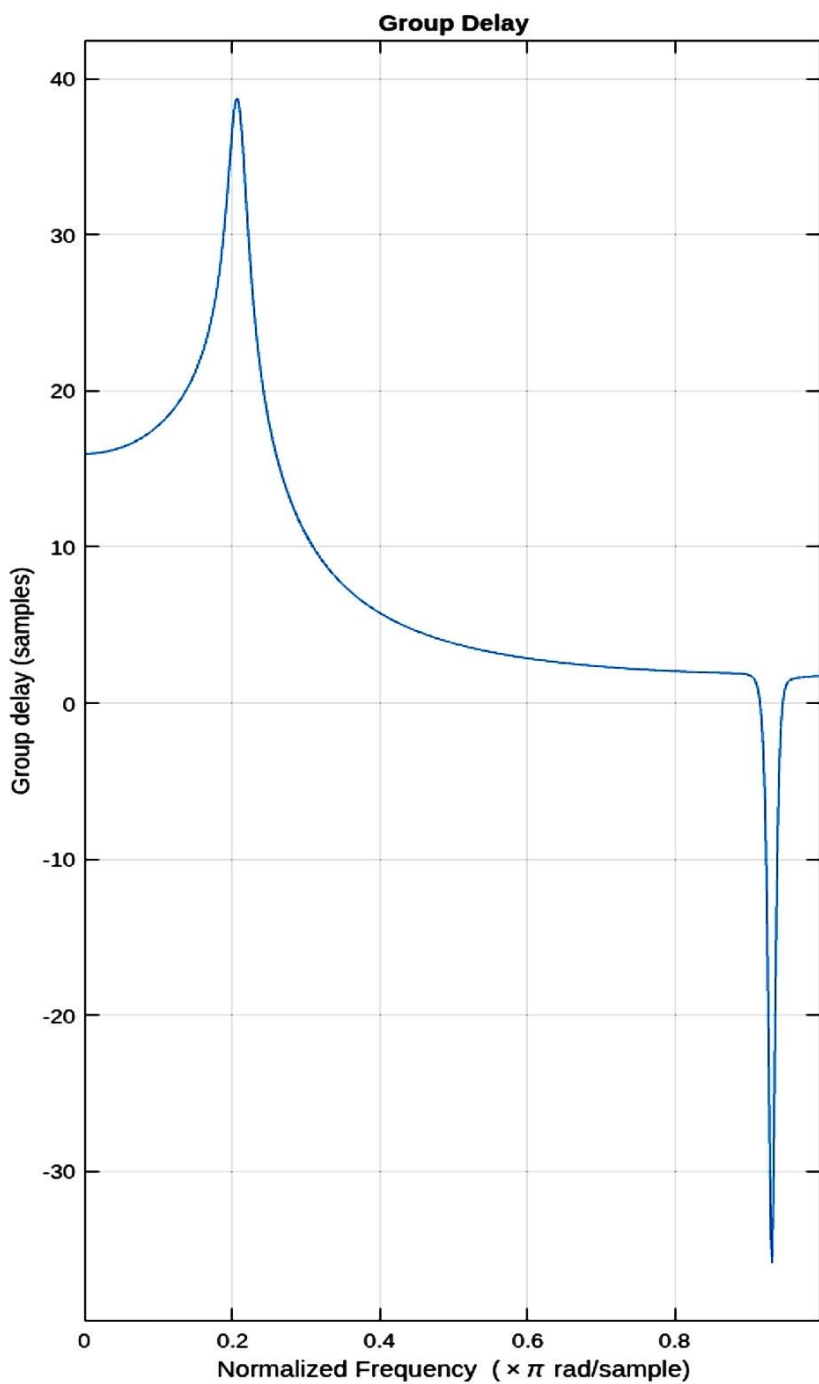


Figures

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Figure 4: Dsp1_1



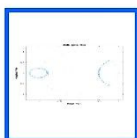
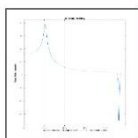
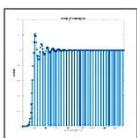
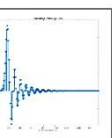
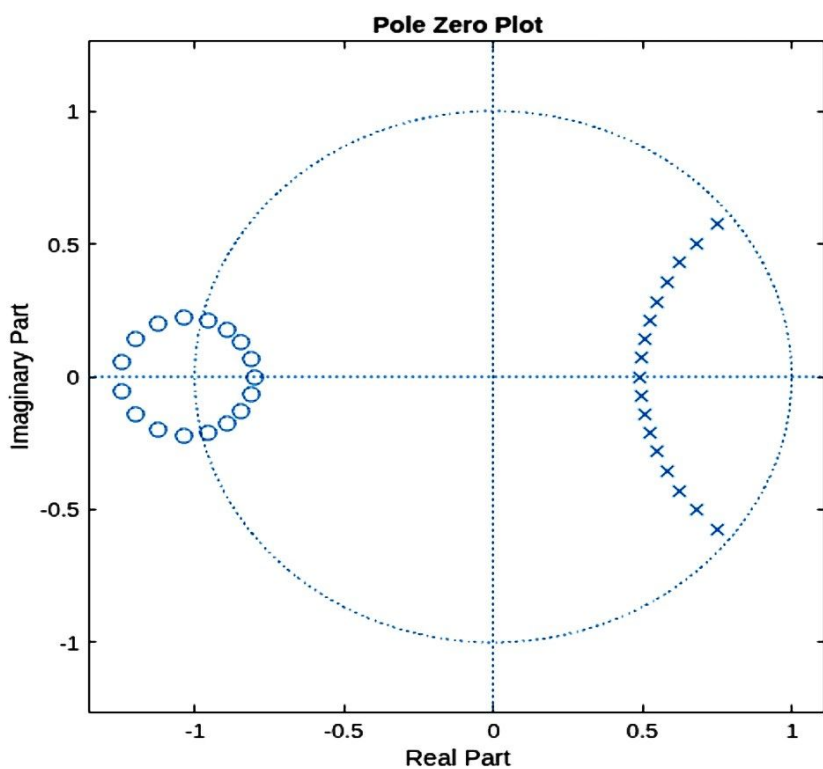


Figures

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Figure 5: Dsp1_1





MATLAB Drive > Dsp1_2.m

```
1 clc;
2 clear;
3 close all;
4
5 % Specifications
6 Fs = 10000;
7 Fp = 3000;
8 Fst = 2000;
9 Rp = 1;
10 Rs = 60;
11
12 % Normalization
13 Wp = Fp/(Fs/2);
14 Ws = Fst/(Fs/2);
15
16 % Order Calculation
17 [N,Wn] = buttord(Wp,Ws,Rp,Rs);
18
19 % High Pass Filter Design
20 [b,a] = butter(N,Wn,'high');
21
22 fprintf('Minimum Filter Order = %d\n',N);
23 fprintf('Cutoff Frequency = %.4f\n',Wn);
24
25 % Frequency Response
26 figure;
27 freqz(b,a);
28 title('Butterworth High Pass Filter');
29
30 % Impulse Response
31 figure;
32 impz(b,a);
33 title('Impulse Response');
34
35 % Step Response
36 figure;
37 stepz(b,a);
38 title('Step Response');
39
40 % Group Delay
41 figure;
42 grpdelay(b,a);
43 title('Group Delay');
44
45 % Pole Zero Plot
46 figure;
47 zplane(b,a);
```



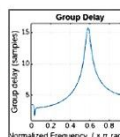
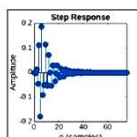
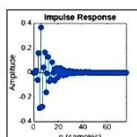
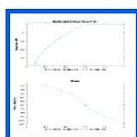
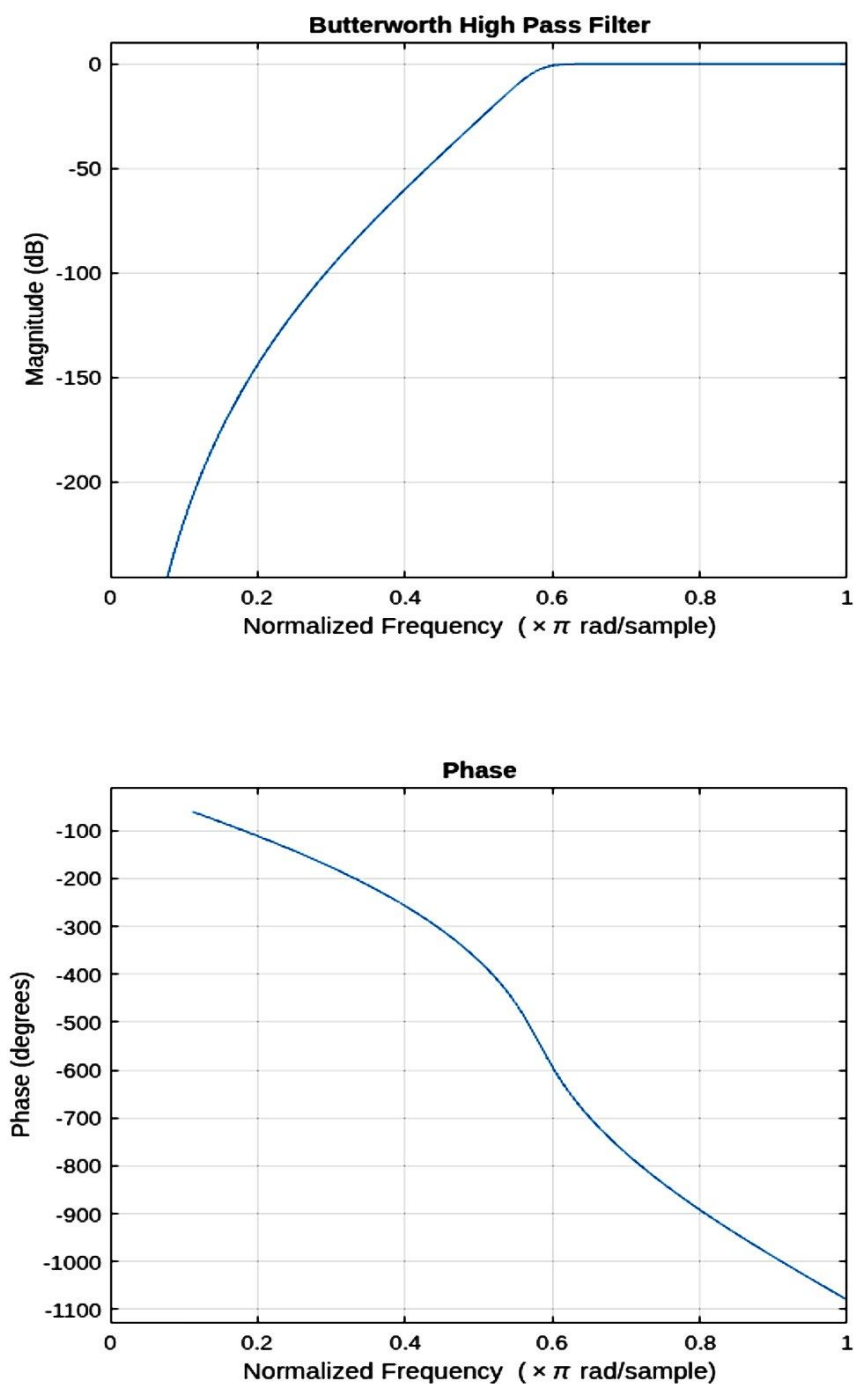


Figures

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Figure 1: Dsp1_2



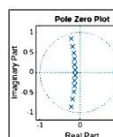
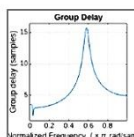
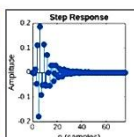
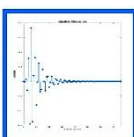
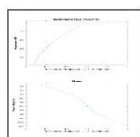
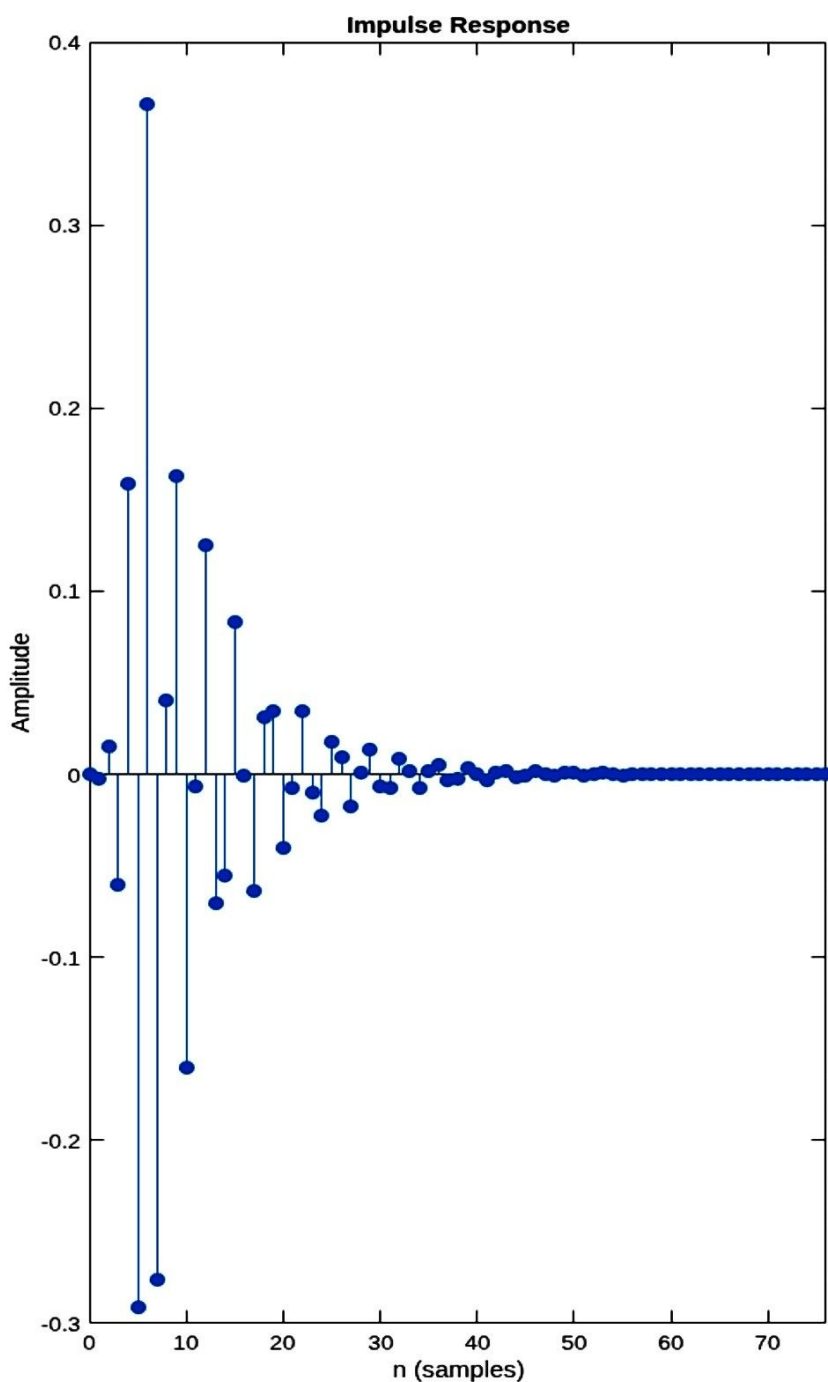


Figures

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Figure 2: Dsp1_2



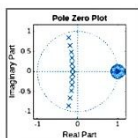
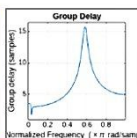
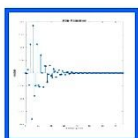
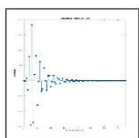
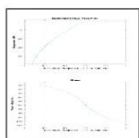
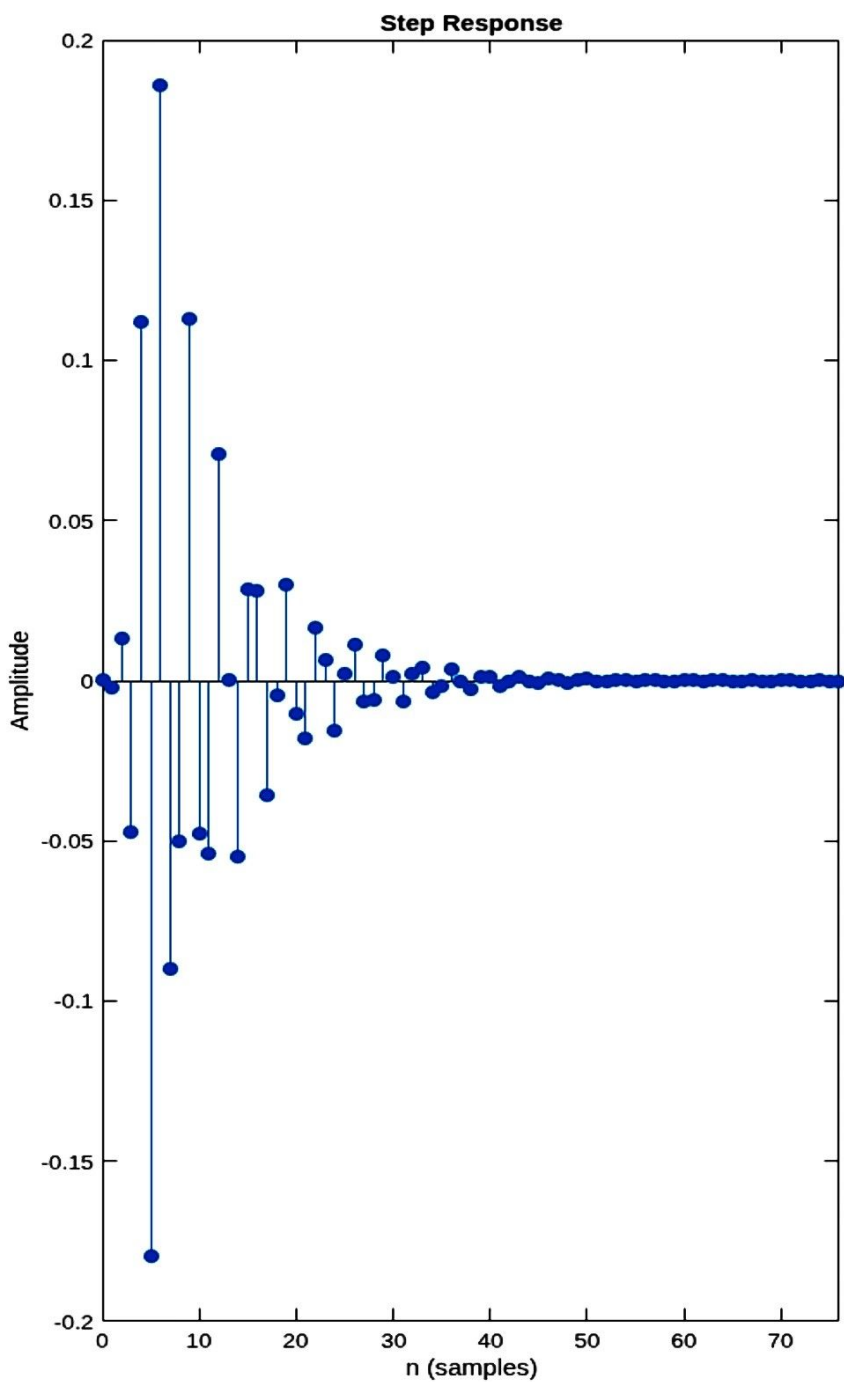


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Figure 3: Dsp1_2



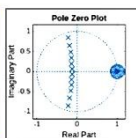
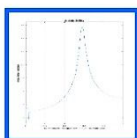
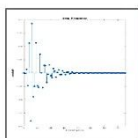
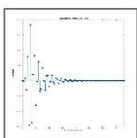
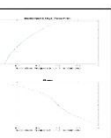
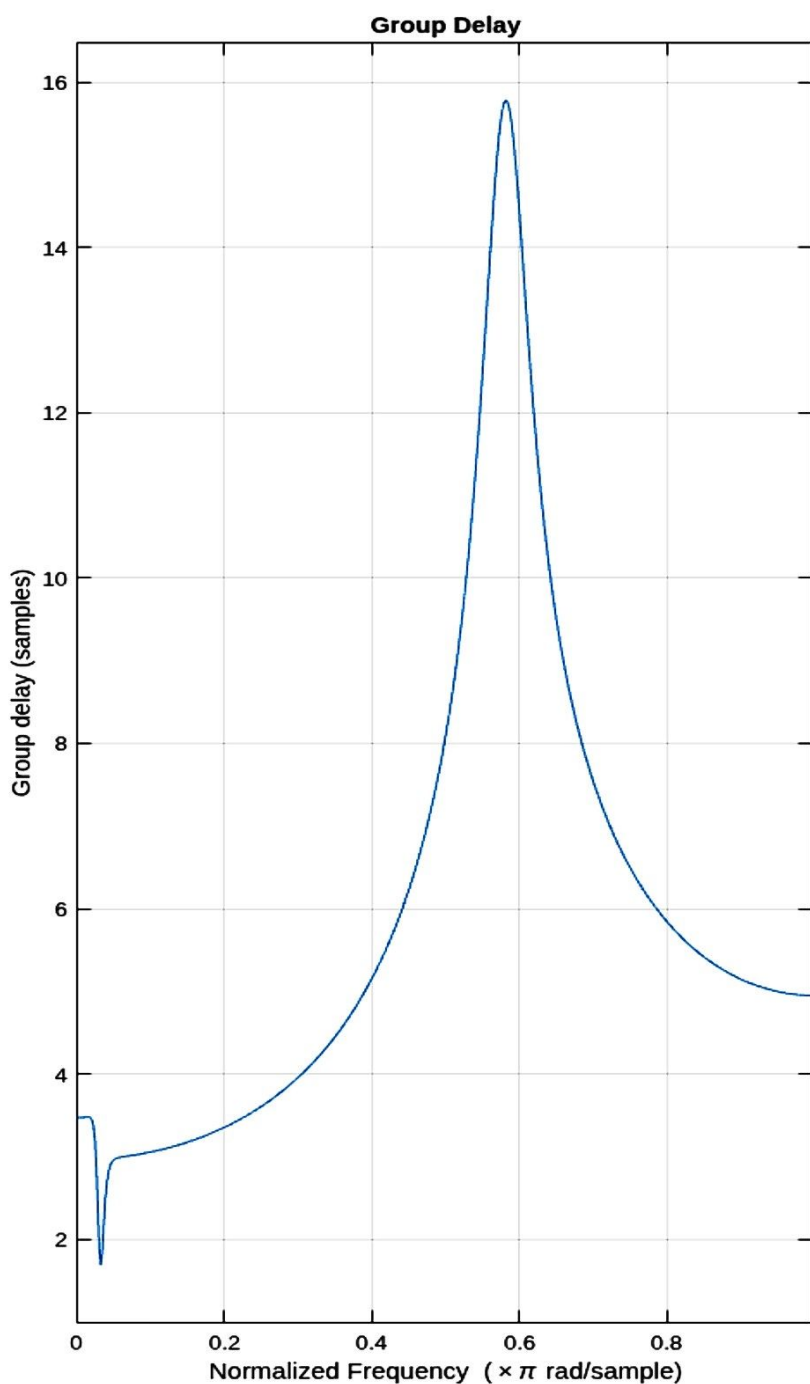


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Figure 4: Dsp1_2





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Figure 5: Dsp1_2

